

NITL\$501- PREPARE SERVER ENVIRONMENT

Section A: Short Answer Questions

1. Introduction to Linux Server Operating Systems

Answer the following questions:

- i. What is a Linux server operating system?
- ii. Identify and describe three main features of Linux server operating systems.
- iii. Describe the key components of a Linux server operating system and explain their functions.
- iv. Explain the main advantages of using a Linux server operating system compared to other server operating systems.
- v. Discuss the common challenges or limitations faced when using Linux server operating systems.
- vi. Identify common use cases of Linux servers in enterprise or organizational environments.

2. Linux Server Requirements

Answer the following questions:

- i. What are the standard hardware requirements for installing a Linux server?
- ii. List the software prerequisites that must be checked before installing a Linux server operating system.
- iii. What are the essential networking requirements when setting up a Linux server?
- iv. Describe the basic security requirements needed to protect a Linux server from unauthorized access and vulnerabilities.
- v. What performance considerations should be made when configuring a Linux server?
- vi. Explain how Linux supports scalability in server environments and outline considerations for scaling up or out.
- vii. Discuss methods that can be implemented to ensure high availability and reliability of Linux servers.

3. Linux Server Setup

Answer the following questions:

- i. What are the minimum hardware specifications required for setting up a basic Linux server?
- ii. Identify three Linux distributions commonly used for servers and highlight their unique features.

iii. List and explain three examples of physical equipment required to establish a Linux server environment.

Section B: Practical Task

Task:

You have been assigned by a school to select and recommend the appropriate tools and equipment needed to set up and manage a Linux server for a small office environment. Identify and justify your choices.

Section C: Multiple Choice Questions

1. **A Linux server operating system is:**
 - a) A system used for personal computing tasks like gaming and web browsing.
 - b) An open-source operating system designed to manage server resources, applications, and services.
 - c) A proprietary software system for managing databases and spreadsheets.
 - d) A system built exclusively for mobile devices.
2. **The main components of a Linux server operating system are:**
 - a) Linux Kernel, Shell, and Package Management System.
 - b) Browser, File Explorer, and Antivirus Program.
 - c) Network Drivers, Email Client, and Social Media Apps.
 - d) Word Processor, Image Editor, and Video Player.
3. **Which of the following are key features of Linux server operating systems?**
 - a) Multitasking, Security, and Open-Source Licensing.
 - b) Video Streaming, File Compression, and Gaming Support.
 - c) Search Engine Optimization, Social Media Integration, and Cloud Storage.
 - d) Proprietary License, Limited Customization, and High Cost.
4. **How does the open-source nature of Linux benefit its development and functionality?**
 - a) It limits contributions to certified developers only.
 - b) It allows anyone to view, modify, and improve the source code, enhancing flexibility and security.
 - c) It prevents customization or future improvements.
 - d) It fixes Linux features permanently without updates.
5. **Which Linux distribution is well-known for stability and is often used in enterprise environments?**
 - a) Ubuntu
 - b) CentOS

- c) Fedora
- d) Arch Linux

Section D: Written Assessment

6. The primary purpose of the **iptables** tool in Linux is:
 - a) Managing software packages
 - b) Configuring network interfaces
 - c) Managing firewall rules
 - d) Monitoring system performance
7. The web server software known for its **lightweight design** and **high performance** is:
 - a) Apache
 - b) Nginx
 - c) IIS
 - d) Tomcat
8. The tool used for **secure remote command-line access** to a Linux server is:
 - a) FTP
 - b) Telnet
 - c) SSH
 - d) HTTP
9. Which tool is commonly used for **monitoring server performance and resource usage**?
 - a) MySQL
 - b) Nagios
 - c) Docker
 - d) Git
10. Which **open-source database management system** is widely used with Linux servers?
 - a) Microsoft SQL Server
 - b) Oracle DB
 - c) MySQL
 - d) IBM Db2

Section E: Fill in the Blanks

Complete the sentences using the correct words from the brackets:

1. A Linux server operating system is designed to manage _____ (hardware resources / software) and provide network services such as _____ (website hosting / file hosting) and database management.
2. The Linux server OS consists of several main components, including the _____ (Kernel / Process), which is the core of the system, and the _____ (File / Shell), which enables user interaction.

Section F: Matching Items

Match each term with its correct description:

Term	Description	Answer
1. Disk Partitioning	A. Software used for configuring and managing firewall rules.	1. ____
2. File System	B. A process that divides a disk into separate, isolated regions.	2. ____
3. Swap Space	C. The area in memory that Linux uses as virtual memory to improve performance.	3. ____
4. LVM (Logical Volume Manager)	D. Used for organizing and storing files on disk.	4. ____
5. Network Interface Configuration	E. A virtualized partitioning system allowing dynamic disk management.	5. ____
6. SSH (Secure Shell)	F. Command-line tool for secure remote access to a server.	6. ____
7. Package Manager	G. System used to install, update, and remove software packages.	7. ____
8. Firewall Configuration	H. Process of configuring IP addresses and gateways.	8. ____

Section G: Scenario-Based Task

Scenario:

Padri Vjeko Centre TSS, a technical and vocational school with over **500 students and 30 trainers**, offers training across four departments — **NIT (Networking and Information Technology)**, **FAD (Fashion and Design)**, **WOT (Wood Technology)**, and **BDC (Building and Construction)**.

To enhance performance, reduce software licensing costs, and improve system reliability, the school plans to **migrate its academic and administrative systems to Ubuntu Linux Server**. The new server environment will host key services such as student management

systems, attendance tracking, file sharing, web hosting for departmental projects, and local intranet access for trainers and students.

Task:

As the **System Administrator** of Padri Vjeko Centre TSS, outline the **steps and considerations** you would take to **install and configure Ubuntu Linux Server** to support the school's operations efficiently. Your plan should cover: